

Nikunj Pradhan

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EDUCATION

Missouri University of Science and Technology

Bachelor of Science, Computer Science, Focus: Data Science

GPA: 4.0/4.0

Aug 2023 - Expected Dec 2026

EXPERIENCE

Full Stack & Data Engineer Co-Op

Jan 2025 – Aug 2025

Hunter Engineering Company

Bridgeton, MO

- Engineered and scaled full-stack applications (FastAPI, React, SQL Server, AWS S3) providing real-time dashboards, data pipelines, and annotation workflows for engineering teams.
- Automated benchmarking workflows by engineering Python CLI tools across AWS and Azure ML pipelines, enabling comparison with vendor models and streamlining team workflow.
- Accelerated distributed multi-GPU training workflows by 66% and development-to-production cycles through close collaboration with R&D teams.
- Developed geometric AI perspective-correction model and integrated into production backend, cutting manual data-cleaning work by 35%.

Backend Engineer

June 2025 – July 2025

Tally Receipts

St. Louis, MO

- Redesigned and migrated backend from MongoDB to AWS DynamoDB, redesigning schemas for efficient querying, scalability, and data lifecycle management.
- Integrated FidelSDK for secure card linking in the frontend, designed business logic for safe card tracking in the database, and implemented webhooks to collect user purchase receipts.
- Refactored and modernized Node.js + Express.js services and integrated AWS Lambda, Amplify, and IAM, streamlining backend logic and infrastructure.
- Built AI-powered receipt categorization service, reducing manual entry effort for users and integrating seamlessly with calendar workflows (Google, Microsoft).

Computer Vision Researcher

Aug 2023 – Dec 2024

Missouri University of Science and Technology

Rolla, MO

- Improved CV model accuracy by 62% through optimization, hyperparameter tuning, and algorithmic refinement using C++ OpenCV and Point Cloud.
- Developed a MATLAB/C++ application simulating 3D triaxial shear tests with deep learning and parallel computing for soil compression modeling, incorporating advanced CV techniques like watershed segmentation and 3D model triangulation via ray tracing.
- Collaborated with other research teams for development, code reviews, and future experimental planning.

PROJECTS

Himalayan Hut | FastAPI, React, PostgreSQL, MinIO, Scikit-learn

- Developed a full-stack movie watchlist platform with secure 2FA login, personalized watchlists, and hybrid recommendations (content-based + ALS collaborative filtering) with per-user caching for accuracy and speed.
- Deployed on Ubuntu using Kubernetes clusters, NGINX reverse proxy, and MinIO for scalable, multi-user media storage and access.

Secure File Server | C++, Crow, OpenSSL, SQLite, MinIO, Docker

- Engineered a self-hosted, end-to-end encrypted file storage and sharing platform with role-based access control, secure uploads/downloads, and metadata management in SQLite.
- Integrated MinIO for scalable object storage and Dockerized deployment for portability, enabling a privacy-first alternative to cloud services like OneDrive/Google Drive.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL, MATLAB, Bash, HTML/CSS

Web Development & APIs: FastAPI, React, Node.js, Express.js, WebSockets, Crow

Cloud & DevOps: AWS (S3, EC2, Lambda), Azure ML, Docker, Kubernetes, Git, CI/CD, Linux/Ubuntu

Databases: MongoDB, PostgreSQL, Microsoft SQL Server, DynamoDB, Firebase, MinIO

Machine Learning & AI: Pytorch, Scikit-learn, ONNX, CNN/CRNN, Transformers, Deep Learning

Data Science Tools: Pandas, NumPy, SciPy, Matplotlib, Jupyter, OpenCV